

CostVision Implementation Blueprint

Secure enterprise deployment inside our network — and integration with CAPEE.
All CAD models, drawings and images stay inside the company. Verified in the code.

**100% CAD stays
internal**

**AI controls BUILT &
tested**

~3-5 weeks remaining

6-phase rollout

PURPOSE OF THIS MEETING

The decision we are asking for today

Approve Phases 1-2: the IT-Security assessment and the architecture design for deploying CostVision inside our network and connecting it to CAPEE. (~5-7 weeks, existing teams, no licence spend.)

Why now

The tool is built, tested (793 automated tests) and proven on live runs — the value is waiting on deployment, not development.

Why it is safe

CAD processing already runs fully inside the server. The private AI routing and the air-gapped switch are now BUILT and tested live.

Why CAPEE wins

CAPEE keeps the workflow; CostVision adds the AI, physics and organisational memory behind it. No tool replacement.

BACKGROUND

What CostVision is — a quick recap

18 cost engines

Machining, casting, injection, PCB, software and more — physics-based, every figure traceable

CAD-to-Cost

Reads STEP/IGES models with a real geometry engine and auto-fills the costing

Photo-to-BOM

Costs a PCB from photographs — detects components and builds the bill of materials

Agentic AI

Learns from every analysis and real quote; suggests, self-corrects and raises findings on its own

Proven, not promised: estimating error cut from 10.9% to 0.3% after learning from 3 real quotes · £512k/yr of pricing issues found autonomously in the live demo · 793 automated tests protect it all.

TECHNOLOGY, SIMPLY EXPLAINED

What the tool is made of — in plain words

Frontend

What users see

The web application in the browser — forms, dashboards, reports. Nothing is installed on laptops; it is served from our own server.

Backend

The engine room

The server application that does the work: runs the 18 cost engines, the AI logic and all calculations. Runs on a standard company virtual machine.

Database

The memory

Where rate libraries, saved costings and the AI knowledge base live. Standard corporate database (PostgreSQL), encrypted, backed up by IT.

CAD engine

The 3D model reader

Specialist geometry software (OCCT — the same core used by CAD vendors) that measures the 3D model: volume, weight, walls, holes. Runs INSIDE the backend.

AI layer

The language brain

The AI model used for vision and language tasks. NOW BUILT: one switchboard controls every AI call — route it to a private endpoint, or turn it fully OFF (air-gapped).

SECURITY

The security requirement — and the key finding

IT-Security requirement: no CAD files, engineering drawings or images may leave the company network. Ever.

The key finding from the code audit: CAD files already never leave the server.

The geometry engine runs inside our backend and processes CAD models in memory — files are not even written to disk, let alone sent anywhere. The only outbound flows are the AI layer (photos and derived summaries) and two optional public feeds. The controls that close this gap are now BUILT: private AI routing and a provable air-gapped switch, tested live in all three modes.

Why you can trust this: this is not a vendor claim — we audited every outbound network call in the platform's source code, line by line, and listed each one. The full inventory is in the written plan.

DATA-FLOW MAP

Where data flows today — verified in the code

STAYS INSIDE — already ✓

CAD / geometry processing

OCCT engine runs in the backend; files processed in memory, never stored

All 18 cost engines

Physics + maths, fully local

Knowledge base & learning loop

Similarity, calibration, autonomous findings — local database

BOM file parsing, exports, reports

Local parsers, local PDF/Excel generation

LEAVES TODAY — controlled ✓

AI layer calls

Photos + CAD-derived summaries. Private routing
BUILT — one setting points every call at our endpoint

Live part pricing (optional, OFF by default)

Part-number text only — silenced by the air-gap switch

News ticker feeds

Public news, no company data — silenced by the air-gap switch

DECISION

Deployment options — our recommendation

OPTION A

Fully air-gapped

Everything on our VMs. AI features off via the built-in AIR-GAPPED switch (already implemented and tested). Deterministic engines fully working.

+ Zero external connections — provable today, switch is built

- Reduced AI capability

OPTION B ★ RECOMMENDED

Private AI, on-prem core

Platform on our VMs. AI calls go to Claude running in OUR OWN cloud tenancy over a private link — no public internet, no data retention.

+ Full capability + data control — routing already BUILT (one setting)

- Requires cloud tenancy sign-off

OPTION C

Public AI API

Platform on-prem but AI calls to the public API.

+ Simplest

- Fails our requirement — photos would traverse a public API

Recommendation: Option B — the software for BOTH A and B is already built; what remains is infrastructure sign-off.

ARCHITECTURE

Target architecture — everything inside our walls

COMPANY INTERNAL NETWORK

Engineers (browser) + CAPEE

single sign-on (Azure AD)

Corporate API Gateway

authentication · rate limits · audit logs

CostVision server (frontend + backend on our VM)

geometry engine (CAD, in-memory) · 18 cost engines · learning loop · report generation

Corporate database (PostgreSQL)

rates · knowledge base · encrypted at rest

Internal AI gateway

ONLY allowed exit · inspected · logged

Zero-trust rules

- Every request authenticated (Azure AD)
- Deny-all egress except the AI gateway
- Role-based access (engineer / lead / admin / CAPEE service)
- CAD pulled from PLM vault, never stored

Our cloud tenancy

Claude on AWS Bedrock / Google Vertex

private link · no public internet · zero data retention

CAD DATA PROTECTION

How CAD data is protected

CAD model opened & measured	Inside our backend (OCCT engine)	✓ Already local — verified
Feature detection (holes, threads, walls)	Inside our backend	✓ Already local — verified
BOM extraction from files	Inside our backend (local parsers)	✓ Already local — verified
Cost calculation & learning	Inside our backend + our database	✓ Already local — verified
Photo analysis & AI narrative	Private AI endpoint (Option B)	→ Routing BUILT — set one value
CAD file storage	Nowhere — processed in memory only	✓ Files never written to disk

Plus: logs never contain CAD payloads; metadata goes to the corporate SIEM; and the "air-gapped" switch is now BUILT & tested — security can PROVE zero egress in a witnessed firewall test.

INTEGRATION

CAPEE + CostVision — better together

CAPEE (stays the front door)

- Costing workflow, approvals, reporting
- System of record — unchanged for users
- Calls CostVision services behind the scenes

requests



AI answers

CostVision (the engine behind it)

- 18 physics cost engines + CAD reading
- AI memory: similar parts, self-calibration
- Autonomous findings for sourcing

What flows over the connection (internal APIs that already exist)

Cost a part → full should-cost with breakdown

CAD file → geometry + auto-filled inputs

New part → similar past parts + suggestions

PO price → feeds the learning loop automatically

Dashboard ← autonomous findings (£/yr)

One shared → rate library & knowledge database

SCOPE OF WORK

What each side needs to change

CostVision — 6 items (2 DONE)

✓ **Private AI routing** — DONE — one setting routes every AI call to our endpoint

✓ **"Air-gapped" switch** — DONE — provably disables all external calls; tested live

3. Azure AD sign-on — replace local logins with company SSO

4. PostgreSQL — move from file database to the corporate DB estate

5. Cost API wrapper — one clean endpoint per commodity for CAPEE

6. Audit hardening — admin audit table + logs to SIEM

CAPEE — small, additive changes

Backend · Small

HTTP client to call CostVision + token handling; hook PO prices into the learning API

User interface · Medium

an "AI insights" panel on the costing screen; a CAD-upload button; a findings widget

Data · Small-Medium

adopt the shared rate library; one-off import of historical quotes to seed the memory

Items 1-2 are built & tested. Remaining critical path: ~3-5 engineering weeks. No architectural surgery on either side.

COMPLIANCE

Security & compliance — how we tick the boxes

Encryption

TLS everywhere in transit; encrypted database at rest; CAD never stored at all

Access control

Azure AD single sign-on; roles mapped to AD groups; CAPEE uses a least-privilege service account

Data-loss prevention

One AI exit door, inspected and logged; firewall denies everything else; witnessed air-gap test (switch built)

Audit trails

Every rate change, knowledge write and admin action attributable to a user; logs to corporate SIEM

Standards mapping: ISO 27001 — covered by the controls above, add to ISMS scope · SOC 2 — inherited from the cloud tenancy (Bedrock/Vertex are certified) · GDPR — minimal personal data, zero-retention AI, EU region · ISO/SAE 21434 — engineering IT tool, out of vehicle scope; supply-chain review (SBOM) in CI.

PLAN

Rollout — six phases, value from week eight



Phase 3 exit gate

Security-witnessed test: firewall blocks all egress, full test suite passes, zero unexpected traffic.

Phase 4 exit gate

A cost engineer completes a real costing from inside CAPEE with AI insights; accuracy dashboard live.

First value

Week ~8: pilot users get similar-part suggestions; sourcing sees the first autonomous findings.

HONEST VIEW

Risks — and how we manage them

Cloud tenancy not approved

Likelihood: Medium

Mitigation: Fall back to Option A (air-gapped) — deterministic engines keep full value today; add a self-hosted vision model later if needed.

Learning depends on logged quotes

Likelihood: Medium

Mitigation: The CAPEE PO-price hook automates it — quotes flow in without anyone changing habits. Plus a one-off historical import to start smart.

Single-team knowledge of the platform

Likelihood: Low-Med

Mitigation: 793 automated tests, written architecture docs, and a named CAPEE-side maintainer trained during Phase 4.

Adoption ("another tool")

Likelihood: Low

Mitigation: Users stay in CAPEE — CostVision works behind the scenes. Nothing new to learn except better answers appearing.

DECISION

Feasibility verdict — and the ask

Secure deployment, CAD fully internal

FEASIBLE — CAD never leaves; AI routing + air-gap switch already BUILT

CAPEE + CostVision integration

FEASIBLE — API-first design; CAPEE stays the front door

Changes required

Head start delivered: 2 of 6 items built — ~3-5 engineering weeks remain

Long-term governance

Rate-library board · monthly findings review · quarterly access & egress audit

The ask today: approve Phase 1 (IT-Security assessment) and Phase 2 (architecture design) — 5-7 weeks, existing teams. The full written plan and security checklist are ready for the security review.